Practice

Part 1

Company ABC contributes to each eligible employee's retirement plan at the rate of 4% of the employee's annual salary. However, to be eligible for this benefit, an employee must have full-time status with two or more years of employment.
There are three retirement contribution possibilities to account for:

- An employee works full-time AND has been employed for two or more years. -> The retirement benefit applies.
- An employee works full time but has NOT been employed for two or more years. ->The retirement benefit does not apply.
- An employee does NOT work full-time. -> The retirement benefit does not apply.

You are required to write the formula that calculates the Retirement Contribution for all the employees.

Moreover, the company also supplies two health plan options:

1. Up to $10K of annual coverage for employees who choose the family plan.
2. Up to $8K of annual coverage for employees who choose the individual plan.

Write the formula to calculate the Health Plan Cost for all the employees.

	B	C	D	E	F	G	H	I
8	Name	Employmt. Status	Health Plan	Salary	Hire Date	# Years Employed	Retirement Contribution	Health Plan Cost
9	Gopnik	part time	family	$45,000	Jan-98	5		
10	Mahfouz	full time	family	$120,000	May-89	13		
11	Bryson	full time	individual	$145,000	Mar-01	2		
12	Peters	full time	individual	$100,000	Nov-00	2		
13	deVries	full time	individual	$115,000	Jul-97	5		
14	Talento	part time	family	$55,000	Aug-95	7		
15	Yang	full time	other plan	$95,000	Apr-99	4		
16	Marks	part time	family	$15,000	May-01	1		
17	Heller	full time	family	$124,000	Oct-00	2		

Part 2

Given below is a voter's registration file for the elections. One cell reference contains the voter's birth date. However, rather than showing a voter was 28, we want to define them as "Young" and so on. Use VLOOKUP to populate the "Segment" column based on the voters' approximate ages.

First	Last	DOB	Age	Segment
Sophia	Collins	9/23/1978	42	
Evelyn	Bennett	2/14/1967	54	
Geneva	Allen	6/6/1990	30	
Silvia	Walsh	5/30/1983	37	
Katrina	Rodriquez	12/23/1998	22	
Sandy	Day	7/24/1973	47	
Dallas	Roberts	6/12/1952	68	
Jamie	Harrington	8/22/1958	62	
Jermaine	Shelton	10/23/1977	43	
Freddie	Baker	5/12/1968	52	

Age	Label
18	New
21	Young
39	Mature
60	Senior

Part 3

You are a data analyst supporting the Academic Affairs Office evaluating grading consistency and student performance patterns for a recent quiz. A random sample of 14 student quiz scores (out of 100) was recorded. Calculate the needed descriptive statistics measures from the raw data to indicate whether each statement below is True or False based and write a brief statistical justification. Rround answers to two decimal places.

- [True / False] The mean score is higher than the median score, indicating that a few high-achieving students are pulling the average upward.
- [True / False] The mode of the dataset indicates that the most frequently occurring score among the students is 93.
- [True / False] The relative variability (CV) of the quiz scores is greater than 25%, reflecting substantial inconsistency in performance relative to the class mean.
- [True / False] The negative skewness coefficient reveals that the score distribution is left-skewed, meaning that most students scored above the mean.
- [True / False] The negative kurtosis coefficient suggests that the distribution is flatter than a normal distribution (platykurtic), with scores more spread out across the range rather than clustered in a sharp central peak.
- [True / False] Based on the empirical z-score criterion ($|z| > 3$), the student who scored 58 is flagged as an extreme outlier.

Part 4

Rename the sheet "Community Information" and answer the following questions:

- What is the number of people living in each of the north, south, east, west?
- How old is the oldest person?
- How old is the youngest person?